S4

In S4, students begin the more formal studying towards the eAssessment. At the end of S5 all students will sit the eAssessmemt at one of two levels: Mathematics is aimed at most students; Extended Mathematics is aimed at the highest achieving students.

At Europa we split the year into 4 classes: Standard; Standard Plus; Extended 2 and Extended 1. You can find further details about this split here.

We decide which students will be in each group at the end of S3, based on performance in the end of year test and also the professional judgement of their S3 teacher. The underlying philosophy is that **we will place students in the group where we believe they will make the most progress, develop the best understanding and achieve the best results**

The structure of the courses is similar, with a few extra sub-topics in each unit for the Extended and Standard Plus groups. Student review and extend their knowledge of various number topics, such as indices and percentages, and then expand their knowledge with new topics in surds (Standard) and logarithms (Extended only). In algebra, students consolidate their knowledge of working with quadratics, and move on to solving quadratic equations, as well as developing algebraic fluency with algebraic fractions (Extended only). This is followed by increasing their knowlwedge of working with volumes and surface areas, and trigonometry, in both 2D and 3D. The year finishes with further development of understanding mathematical models, first developing their understanding of linear models, and then moving on to quadratic models.

S4 students will also begin to develop their more formal investigation and pattern spotting skills, a vital part of being a mathematician, which is assessed explicitly in the eAssessment. Student can practice their skills at finding the nth term of sequences, as well as the investigation structure.

S4 sees students develop their algebraic skills alongside modelling skills which are so vital in b ⓘ ʒ able to answer a wide variety of mathematical problems.

In this unit students will learn about:

- Linear sequences and finding the nth term
- Combinations of linear sequences
- Quadratic sequences and finding the nth term
- Geometric sequences and finding the nth term
- Formal notation for sequences (Extended Only)
- The process and layout of a Mathematical Investigation

Unit 2 - Linear Functions

In this unit students will learn about:

- The midpoint of two points
- The distance between two points
- Finding the gradient of lines
- The equation of a straight line in the forms y=mx+c and ax+by=d
- Working with parallel lines
- Working with perpendicular lines (Extended Only)
- Solving pairs of linear simultaneous equations
- Finding perpendicular bisectors and shortest distances (Extended Only)

Unit 3 - Measures

In this unit students will learn about:

- Converting between metric units of area and volume
- Volumes and surface areas of 3D shapes
- Using trigonometry in 3D shapes
- Calculating with similar shapes
- The idea of congruence

In this unit students will learn about:

- The different types of number and their notations
- Sets and set notation
- Laws of indices
- Working with bounds (Extended Only)
- Absolute values
- Surds
- Logarithms (Extended Only)

Unit 5 - Algebra

In this unit students will learn about:

- Solving quadratic equations by factorising and by quadratic formula
- Direct and inverse proportion
- Rearranging formulae
- Algebraic Fractions (Extended Only)
- Solving linear inequalities and representing these on the number line
- Solving problems on linear programming involving linear inequalities in the Cartesian plane (Extended Only)
- Solving quadratic inequalities (Extended Only)
- Simple Proof (Extended Only)

In this unit students will learn about:

- What functions are
- Graphing functions
- Composite functions (Extended Only)
- Inverse functions (Extended Only)
- Domain and range (Extended Only)
- Reciprocal functions
- Exponential functions
- Trigonometric functions
- Circles (Extended Only)
- Transforming functions (Extended Only)

Unit 7 - Geometry

In this unit students will learn about:

- Circle theorems
- Working with arcs and sectors of circles
- Transformations of shapes (reflection, rotations, translations and enlargements)
- Vectors (Extended Only)

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